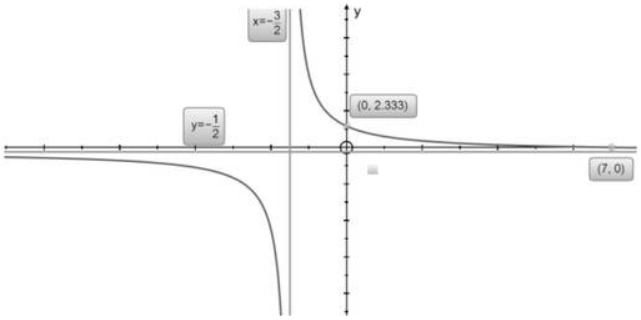
Type MS · The quotient rule: differentiating fractions

Answer 1 of 16 · Question 10 part d · 3 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part d in this mark scheme extract

10 (a)	(i) $x=-\frac{3}{2}$ (ii) $y=-\frac{1}{2}$	B1 B1 [2]
(b)	$(7,0)$ $\left(0,\frac{7}{3}\right)$	B1 B1 [2]
(c)		B1ft curve B1ft asymptotes B1ft (intersections with x- and y-axes) [3]
(d)	$\frac{-1(2x+3)-2(7-x)}{(2x+3)^2}=\left[\frac{-17}{(2x+3)^2}\right]$ $\frac{-1\left(2\times\frac{1}{2}+3\right)-2\left(7-\times\frac{1}{2}\right)}{\left(2\times\frac{1}{2}+3\right)^2}=-\frac{17}{16}$ ALT – product rule $-1(2x+3)^{-1}+(7-x)(-1)(2)(2x+3)^{-2}=\left[\frac{-17}{(2x+3)^2}\right]$ $-1\left(2\times\frac{1}{2}+3\right)^{-1}+\left(7-\frac{1}{2}\right)(-1)(2)\left(2\times\frac{1}{2}+3\right)^{-2}=-\frac{17}{16}$	M1 M1A1* cso [3] M1 M1A1* cso

Type MS · The quotient rule: differentiating fractions

Answer 1 of 16 · Question 10 part d · 3 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part d in this mark scheme extract

(e)	$-\frac{17}{(2x+3)^2}=-\frac{17}{16}$	M1
	$(2x+3)^2=16 \Rightarrow 2x+3=\pm\sqrt{16} \Rightarrow x=0.5 \text{ or } -3.5$	dM1
	$y=\frac{7--3.5}{2\times-3.5+3}\left(=-\frac{21}{8}\right)$	ddM1
	$\left(-\frac{7}{2},-\frac{21}{8}\right)$	A1
	$y--\frac{21}{8}=-\frac{17}{16}\left(x--\frac{7}{2}\right)$	M1A1
	$34x+32y+203=0$	A1 [7]
		Total 17 marks

Type MS · The quotient rule: differentiating fractions

Answer 1 of 16 · Question 10 part d · 3 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part d in this mark scheme extract

Part	Mark	Additional Guidance
For parts (a) and (b) look for their responses in the body of the question. In part (a), if the responses are not labelled (i) and (ii) accept them in order.		
(a)	(i) B1	For $x=-\frac{3}{2}$ oe
	(ii) B1	For $y=-\frac{1}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x=7$ and/or $y=\frac{7}{3}$ given
(c)	B1ft	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(d)	M1	Attempt the quotient rule. Numerator must be of the form $A.(2x+3)-B.(7-x)$, $B>1$. Denominator must be of the form $(2x+3)^2$
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x+3)^{-1}+d(7-x)(2x+3)^{-2}$ for constants c, d.
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
(e)	M1	For equating the gradient of the tangent to their $\frac{dy}{dx}$ from (d)
	dM1	For rearranging to obtain $(2x+3)^2=16$ and attempt to solve or for rearrangement to a 3TQ and method to solve the 3TQ. See general guidance for acceptable methods to solve.

Type MS · The quotient rule: differentiating fractions

Answer 1 of 16 · Question 10 part d · 3 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part d in this mark scheme extract

	ddM1	For substitution of their x into the original equation to obtain y. This must **not** be $x=\frac{1}{2}$ This M mark is dependent on the **two** previous M marks
	A1	For correct coordinate of $P\left(-\frac{7}{2},-\frac{21}{8}\right)$
	M1	For substitution of their coordinates of P and the gradient $-\frac{17}{16}$ to obtain an equation of a straight line using any valid method. If $y=mx+c$ is used a value of c must be found **and** an equation needs to be formed. For example; $'-\frac{21}{8}'=-\frac{17}{16}\times'-\frac{7}{2}'+c\Rightarrow c=-\frac{203}{32}\Rightarrow y=-\frac{17}{16}x-\frac{203}{32}$ [
	A1	For any correct equation of the tangent with the correct values of x and y in any form.
	A1	For the correct equation of tangent given in the required form. $34x+32y+203=0$

Type MS · The quotient rule: differentiating fractions

Answer 1 of 16 · Question 10 part d · 3 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 2 of 16 · Question 8 part a · 5 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

8(a)	$2xy+5y=e^x \quad y=\dfrac{e^x}{(2x+5)}$	
	$\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$	M1A1A1
	$\dfrac{dy}{dx}=\dfrac{e^x}{(2x+5)}\times\dfrac{(2x+5-2)}{(2x+5)}=\dfrac{y(2x+3)}{(2x+5)}$ *	M1A1 (5)
(b)	$x=0\Rightarrow \dfrac{dy}{dx}=\dfrac{5-2}{5^2}=\dfrac{3}{25}$	M1A1 (2)
ALT	$x=0\Rightarrow y=\dfrac{1}{5},\ \dfrac{dy}{dx}=\dfrac{1}{5}\times\dfrac{3}{5}=\dfrac{3}{25}$	
(c)	$x=0\Rightarrow y=\dfrac{e^0}{(2\times0+5)}=\dfrac{1}{5}$	M1(Award if seen in (b) and **used** in (c))
	$y-\dfrac{1}{5}=-\dfrac{25}{3}x$	M1
	$125x+15y-3=0$	A1 (3) [10]

Part	Mark	Notes
(a)		$2xy+5y=e^x \Rightarrow y=\dfrac{e^x}{(2x+5)}$
	M1	For attempting Quotient Rule • Both terms must be differentiated correctly $e^x\Rightarrow e^x \quad 2x+5\Rightarrow 2$ • There must be two terms subtracted in the numerator either way around • The denominator must the denominator squared. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$
	A1	For $e^x(2x+5)$ or $2e^x$
	A1	For a fully correct differentiated expression. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$

Type MS · The quotient rule: differentiating fractions

Answer 2 of 16 · Question 8 part a · 5 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

	M1	Subs in $y=\frac{e^x}{(2x+5)}$ as a common factor $\frac{dy}{dx}=\frac{e^x}{(2x+5)}\times\frac{(2x+5-2)}{(2x+5)}=\frac{y(2x+5-2)}{(2x+5)}$	Subs in $e^x=y(2x+5)$ and factorises $\frac{dy}{dx}=\frac{y(2x+5)(2x+5)-2y(2x+5)}{(2x+5)^2}$
	A1	For the correct answer with no errors. **Note this is a given answer.** $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
	ALT – uses implicit differentiation on $2xy+5y=e^x$		
	M1	$2\left(y+x\frac{dy}{dx}\right)+5\frac{dy}{dx}=e^x$	
	A1	Takes out $\frac{dy}{dx}$ as a common factor $\frac{dy}{dx}(2x+5)=e^x-2y$	
	A1	For a fully correct differentiated expression as below. $\frac{dy}{dx}=\frac{e^x-2y}{(2x+5)}$	
	M1	For separating the fraction, taking out y as a common factor and attempting to form a single fraction $\frac{dy}{dx}=\frac{e^x}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5)}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5-2)}{2x+5}$	
	A1	For the correct answer with no errors. $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
(b)	M1	For substituting $x=0$ into $\frac{dy}{dx}=\frac{e^x(2x+5)-2e^x}{(2x+5)^2}=\frac{e^0(2\times0+5)-2e^0}{(2\times0+5)^2}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
	ALT		
	M1	When $x=0\Rightarrow y=\frac{1}{5}$, $\frac{dy}{dx}=\frac{1}{5}\times\frac{3}{5}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
(c)	M1	$x=0\Rightarrow y=\frac{e^0}{(2\times0+5)}=\frac{1}{5}$ $x=0\Rightarrow y=\frac{1}{5}$ Award if seen in (b) and **used** in (c) **This is a B mark in Epen.**	
	M1	Inverts the gradient found in (b) and forms equation of the normal. ft their value of y $y-\frac{1}{5}=-\frac{25}{3}x$	
	A1	Equation of line is given in the required form. $125x+15y-3=0$	

Type MS · The quotient rule: differentiating fractions

Answer 2 of 16 · Question 8 part a · 5 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	$6^2 + x^2 - 12^2 \quad x^2 - 108$	

Type MS · The quotient rule: differentiating fractions

Answer 3 of 16 · Question 8 part b · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

8(a)	$2xy+5y=e^x \quad y=\dfrac{e^x}{(2x+5)}$	
	$\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$	M1A1A1
	$\dfrac{dy}{dx}=\dfrac{e^x}{(2x+5)}\times\dfrac{(2x+5-2)}{(2x+5)}=\dfrac{y(2x+3)}{(2x+5)}$ *	M1A1 (5)
(b)	$x=0\Rightarrow \dfrac{dy}{dx}=\dfrac{5-2}{5^2}=\dfrac{3}{25}$	M1A1 (2)
ALT	$x=0\Rightarrow y=\dfrac{1}{5},\ \dfrac{dy}{dx}=\dfrac{1}{5}\times\dfrac{3}{5}=\dfrac{3}{25}$	
(c)	$x=0\Rightarrow y=\dfrac{e^0}{(2\times0+5)}=\dfrac{1}{5}$	M1(Award if seen in (b) and **used** in (c))
	$y-\dfrac{1}{5}=-\dfrac{25}{3}x$	M1
	$125x+15y-3=0$	A1 (3) [10]

Part	Mark	Notes
(a)		$2xy+5y=e^x \Rightarrow y=\dfrac{e^x}{(2x+5)}$
	M1	For attempting Quotient Rule • Both terms must be differentiated correctly $e^x\Rightarrow e^x \quad 2x+5\Rightarrow 2$ • There must be two terms subtracted in the numerator either way around • The denominator must the denominator squared. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$
	A1	For $e^x(2x+5)$ or $2e^x$
	A1	For a fully correct differentiated expression. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$

Type MS · The quotient rule: differentiating fractions

Answer 3 of 16 · Question 8 part b · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

	M1	Subs in $y=\frac{e^x}{(2x+5)}$ as a common factor $\frac{dy}{dx}=\frac{e^x}{(2x+5)}\times\frac{(2x+5-2)}{(2x+5)}=\frac{y(2x+5-2)}{(2x+5)}$	Subs in $e^x=y(2x+5)$ and factorises $\frac{dy}{dx}=\frac{y(2x+5)(2x+5)-2y(2x+5)}{(2x+5)^2}$
	A1	For the correct answer with no errors. **Note this is a given answer.** $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
	ALT – uses implicit differentiation on $2xy+5y=e^x$		
	M1	$2\left(y+x\frac{dy}{dx}\right)+5\frac{dy}{dx}=e^x$	
	A1	Takes out $\frac{dy}{dx}$ as a common factor $\frac{dy}{dx}(2x+5)=e^x-2y$	
	A1	For a fully correct differentiated expression as below. $\frac{dy}{dx}=\frac{e^x-2y}{(2x+5)}$	
	M1	For separating the fraction, taking out y as a common factor and attempting to form a single fraction $\frac{dy}{dx}=\frac{e^x}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5)}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5-2)}{2x+5}$	
	A1	For the correct answer with no errors. $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
(b)	M1	For substituting $x=0$ into $\frac{dy}{dx}=\frac{e^x(2x+5)-2e^x}{(2x+5)^2}=\frac{e^0(2\times0+5)-2e^0}{(2\times0+5)^2}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
	ALT		
	M1	When $x=0\Rightarrow y=\frac{1}{5}$, $\frac{dy}{dx}=\frac{1}{5}\times\frac{3}{5}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
(c)	M1	$x=0\Rightarrow y=\frac{e^0}{(2\times0+5)}=\frac{1}{5}$ $x=0\Rightarrow y=\frac{1}{5}$ Award if seen in (b) and **used** in (c) **This is a B mark in Epen.**	
	M1	Inverts the gradient found in (b) and forms equation of the normal. ft their value of y $y-\frac{1}{5}=-\frac{25}{3}x$	
	A1	Equation of line is given in the required form. $125x+15y-3=0$	

Type MS · The quotient rule: differentiating fractions

Answer 3 of 16 · Question 8 part b · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$6^2 + x^2 - 12^2 \quad x^2 - 108$	

Type MS · The quotient rule: differentiating fractions

Answer 4 of 16 · Question 3 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
3 (a)	$\frac{dy}{dx} = 2e^{2x}\sqrt{5x-3} + \frac{5e^{2x}}{2\sqrt{5x-3}}$	M1 A1 A1 (3)
(b)	$\frac{dy}{dx} = \frac{3x^2\cos 3x - x^3(-3\sin 3x)}{(\cos 3x)^2}$	M1 A1 A1 (3)
		Total 6 marks

Type MS · The quotient rule: differentiating fractions

Answer 4 of 16 · Question 3 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 5 of 16 · Question 8 part whole · 5 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

		$\pm\frac{b}{x^3}+8\ b\neq 0$
	A1	Correctly substitutes $x = 3$ into the correct second derivative or argues the correct second derivative is > 0 and draws a conclusion. It is not necessary to see an evaluation, but the evaluation must be correct if carried out. Second derivative **must be stated to be > 0 or positive.** As always with A1, must have gained all previous marks. **Note: it is highly unusual to have a final A1, when any of the preceding marks have been 0. The structure of this question means, this is possible, if they either don't find the minimum area or calculate it incorrectly.** **So for the final 4 marks dM1 A0 M1 A1 and dM0 A0 M1 A1**

Type MS · The quotient rule: differentiating fractions

Answer 5 of 16 · Question 8 part whole · 5 marks

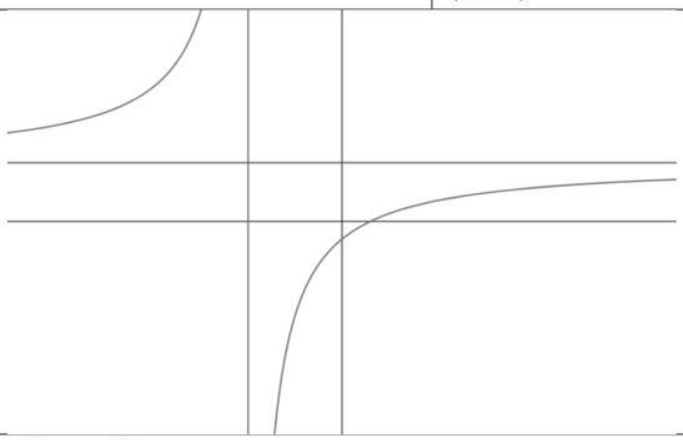
2025 JUN · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks
8	$3e^{3x}(9x+2)^{\frac{1}{3}}+\frac{1}{3}.9.e^{3x}(9x+2)^{-\frac{2}{3}}$	M1 A1 A1
	$3e^{3}(9+2)^{\frac{1}{3}}+\frac{1}{3}.9.e^{3}(9+2)^{-\frac{2}{3}}$	M1
	$=146$	A1 [5]
		Total 5 marks

Type MS · The quotient rule: differentiating fractions

Answer 6 of 16 · Question 10 part d · 4 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 4 · find part d in this mark scheme extract

10(a) Mark (i) and (ii) together	$-\frac{1}{4}=-\frac{2}{b}\rightarrow b=8$ *	B1*cso
	$-\frac{4}{3}a+8=0$ oe or $-\frac{b}{a}=-\frac{4}{3}$	M1
	$a=6$	A1 [3]
(b)	$y=\frac{5}{"6"}$	B1ft [1]
(c)	$\left(\frac{2}{5},0\right)$	B1 [1]
(d)	$\frac{5(6x+8)-6(5x-2)}{(6x+8)^2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion ALT using product rule $(5x-2)(6x+8)^{-1}$ $5(6x+8)^{-1}+(-1)(6)(5x-2)(6x+8)^{-2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion	M1 dM1 A1 A1 [4]
(e)		B1 (curve) B1ft (asymptotes) B1ft (intersections with axes) [3]
(f)	$\frac{13}{100}=\frac{52}{(6x+8)^2}$ oe	M1
	$(6x+8)^2=400$ **$3x^2+8x-28=0$ oe**	dM1
	$x=2, y=\frac{2}{5}$ $x=-\frac{14}{3}, y=\frac{19}{15}$ oe	A1 A1
	$\sqrt{\left(("2")-\left("-\frac{14}{3}"\right)\right)^2+\left(\left("\frac{2}{5}"\right)-\left("\frac{19}{15}"\right)\right)^2}$	dM1
	6.72	A1 [6]
		Total 18 marks

Type MS · The quotient rule: differentiating fractions

Answer 6 of 16 · Question 10 part d · 4 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 4 · find part d in this mark scheme extract

Part	Mark	Additional Guidance
(a) (i) and (ii)	B1*cso	For a minimum of the steps shown in the MS to find b, no errors.
	M1	For correct substitution of $x=-\frac{4}{3}$ and $b=8$ into $ax+b=0$
	A1	Correct a.
		Students can complete this in a different order $-\frac{4}{3}a+8=0 \Rightarrow a=6$ $-\frac{b}{a}=-\frac{4}{3}\left(\text{or } -\frac{b}{6}=-\frac{4}{3}\right)\Rightarrow b=8$ B1 M1 A1
(b)	B1ft	Follow through their value of a. $y=\frac{5}{"their 6"}$
(c)	B1	$\left(\frac{2}{5},0\right)$ allow $x=\frac{2}{5}$ $y=0$ $x=\frac{2}{5}$ alone is insufficient
(d)	M1	Attempt the quotient rule. Numerator must be the difference of two terms of the form $A\times(ax+8)-B\times(5x-2)$ or $B\times(5x-2)-A\times(ax+8)$, A and $B>1$. Denominator must be of the form $(ax+8)^2$ $a\neq 0$ Allow follow through of their value of a (general principle of method mark).
	dM1	Either term on the numerator fully correct (either way round), dependent on previous method mark, doesn't need to be simplified. Denominator must be of the form $(ax+8)^2$ Allow follow through of their value of a (general principle of method mark).
	A1	$\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ eg $\frac{13}{(3x+4)^2}$ A simplified expression (denominator expanded acceptable) Dependent on both previous method marks – question states must use calculus, so some method must be shown.
	A1	$\frac{52}{(6x+8)^2}$ and correct conclusion, for example, (the numerator is a positive number) eg the denominator is always positive eg the $(ax+8)^2>0$ and therefore (the fraction) is always positive. No incorrect work for this mark eg don't accept incorrectly simplified gradient.
ALT	M1	Attempt the product rule. $C\times(ax+8)^{-1}+(5x-2)(-(1))(D)\ (ax+8)^{-2}$, C and $D>1$. Allow follow through of their value of a (general principle of method mark).
	M1	Either term correct (either way round), dependent on previous method mark, doesn't need to be simplified
	A1 A1	As main scheme

Type MS · The quotient rule: differentiating fractions

Answer 6 of 16 · Question 10 part d · 4 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 4 · find part d in this mark scheme extract

(e)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes $x = -\frac{8}{\text{their } a}$, $y = \frac{5}{\text{their } a}$, ft their value for a, labelled as equations or clearly passing through the correctly labelled numbers on the axis. There must be one section of a negative reciprocal curve present, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through $\left(0, -\frac{1}{4}\right)$ and $\left(\frac{2}{5}, 0\right)$ marked clearly on the graph as coordinates or crossing points on the axis. Ignore any other branch/curve/line present. Allow follow through only of their $(c, 0)$ $c \neq 0$ from part (c)
(f)	M1	Sets their differentiated function from part (d) = $\frac{13}{100}$ Allow any function = $\frac{13}{100}$ coming from working in part (d) or re-started work to find the derivative. The function may not be the original function.
	dM1	For correctly rearranging an equation of the form $\frac{13}{100} = \frac{Q}{(cx+d)^2}$ to $(cx+d)^2 = R$ using their a, or to the form $px^2 + qx + r = 0$ $p, q, r \neq 0$ with 2 out of 3 of p, q, r correct. leading to $x =$
	A1	For $x = 2, y = \frac{2}{5}$ or $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	A1	For $x = 2, y = \frac{2}{5}$ and $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	dM1	For correct use of the formula to find the length of a line, using their values for x and y. Dependent on **first** method mark only.
	A1	For awrt 6.72

Type MS · The quotient rule: differentiating fractions

Answer 6 of 16 · Question 10 part d · 4 marks

2025 JUN · Paper 02 (regional R) · extract 4 of 4 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 7 of 16 · Question 7 part b · 4 marks

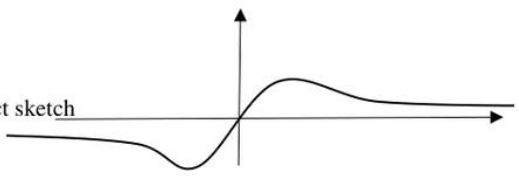
2021 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	$\frac{dy}{dx} = \frac{(x^2+4)-x(2x)}{(x^2+4)^2}$	M1
	$= \frac{4-x^2}{(x^2+4)^2}$	A1
	$\frac{dy}{dx} = 0 \Rightarrow 4-x^2 = 0$	M1
	So $\left(2, \frac{1}{4}\right)$ and $\left(-2, -\frac{1}{4}\right)$	A1 A1 (5)
(b)	$\frac{d^2y}{dx^2} = \frac{-2x(x^2+4)^2 - 4x(4-x^2)(x^2+4)}{(x^2+4)^4}$	M1
	$\frac{d^2y}{dx^2} = \frac{-2x^3-8x-16x+4x^3}{(x^2+4)^3}$	M1
	$\frac{d^2y}{dx^2} = \frac{2x^3-24x}{(x^2+4)^3}$	M1
	$\frac{d^2y}{dx^2} = \frac{2x(x^2-12)}{(x^2+4)^3}$ *	A1 cso (4)
(c)	When $x = 2$ $\left[\frac{d^2y}{dx^2} = -\frac{1}{16}\right]$ When $x = -2$ $\left[\frac{d^2y}{dx^2} = \frac{1}{16}\right]$	
	$\frac{d^2y}{dx^2} < 0$ so maximum $\frac{d^2y}{dx^2} > 0$ so minimum	M1 A1ft (2)
		Total 11 marks

Type MS · The quotient rule: differentiating fractions

Answer 7 of 16 · Question 7 part b · 4 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

	A1	For the correct coordinates of either $\left(2,\frac{1}{4}\right)$ or $\left(-2,-\frac{1}{4}\right)$
	A1	For both correct coordinates $\left(2,\frac{1}{4}\right)$ and $\left(-2,-\frac{1}{4}\right)$
(b)	**M1**	For an attempt at Quotient rule on their $\frac{dy}{dx}=\frac{(x^2+4)-x(2x)}{(x^2+4)^2}$ which must be as a minimum: $\frac{ax^2+bx+c}{(x^2+4)^2}$ where a, b and c are constants and $a,c\neq 0$ The definition of an attempt is that there must be a minimal attempt to differentiate the numerator and denominator and the correct formula applied. $(x^2+4)^2$ must differentiate to $ax(x^2+4)$, the denominator must be $(x^2+4)^4$ Allow the terms in the numerator to be the wrong way around, but the terms must be subtracted. Apply General Guidance for an attempt at differentiation on ax^2+bx+c. $\frac{d^2y}{dx^2}=\frac{-2x(x^2+4)^2-4x(4-x^2)(x^2+4)}{(x^2+4)^4}$
	M1	For cancelling through by (x^2+4) $\frac{d^2y}{dx^2}=\frac{-2x(x^2+4)-4x(4-x^2)}{(x^2+4)^3}$
	M1	For simplifying the numerator to achieve as a minimum $\frac{d^2y}{dx^2}=\frac{ax^3+bx}{(x^2+4)^3}$ where a and b are constants
	A1 cso	For obtaining the answer as given with no errors. $\frac{d^2y}{dx^2}=\frac{2x(x^2-12)}{(x^2+4)^3}$ *
(c)	**M1**	Substitutes either their 2 or their – 2 into their $\frac{d^2y}{dx^2}$ Note: When $x=2\left[\frac{d^2y}{dx^2}=-\frac{1}{16}\right]$ and when $x=-2\left[\frac{d^2y}{dx^2}=\frac{1}{16}\right]$
	A1ft	For the conclusion $\frac{d^2y}{dx^2}<0$ so maximum $\frac{d^2y}{dx^2}>0$ so minimum
	ALT – tests gradient or sight of a sketch	
	M1	Tests gradient on either side of one the turning points (their 2 or their – 2) using their $\frac{dy}{dx}$ or a correct sketch 
	A1ft	For the correct conclusion

Type MS · The quotient rule: differentiating fractions

Answer 7 of 16 · Question 7 part b · 4 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 8 of 16 · Question 5 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

5 (a)	$4e^{4x}(6x+2)^{\frac{3}{2}}+\frac{3}{2}.6.e^{4x}(6x+2)^{\frac{1}{2}}=\left[4e^{4x}(6x+2)^{\frac{3}{2}}+9e^{4x}(6x+2)^{\frac{1}{2}}\right]$ $e^{4x}\sqrt{6x+2}(24x+17)$	M1 A1 A1 dM1 A1 [5]
(b)	$\frac{3\cos 3x(2x-4)^3-\sin 3x.2.3.(2x-4)^2}{\left[(2x-4)^3\right]^2}=\left[\frac{3\cos 3x(2x-4)^3-6\sin 3x(2x-4)^2}{(2x-4)^6}\right]$ OR $\frac{3\cos 3x(2x-4)^3-\left(24x^2-96x+96\right)\sin 3x}{(2x-4)^6}$ ALT using product rule $\sin 3x(2x-4)^{-3}$ $3\cos 3x(2x-4)^{-3}+\sin 3x(-3)(2)(2x-4)^{-4}=\left[3\cos 3x(2x-4)^{-3}-6\sin 3x(2x-4)^{-4}\right]$	M1 A1 A1 M1 A1 A1 [3]
	Total	**8 marks**

Type MS · The quotient rule: differentiating fractions

Answer 8 of 16 · Question 5 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Use of product rule to give an expression of the form $pe^{4x}(6x+2)^{\frac{3}{2}}+qe^{4x}(6x+2)^{\frac{1}{2}}$ where $p = 1$ or 4, $q > 1$. There must be a + sign between terms.
	A1	Either term correct. Note that simplification is not required for this mark.
	A1	Both terms correct. Note that simplification is not required for this mark.
	dM1	Obtains an expression of the form $e^{4x}\left(\sqrt{6x+2}\right)(Ax+B)$ where $A, B \neq 0$ You **must** check their working here that it follows from their attempt at product rule. There must be an intermediate step seen as follows for example: $4e^{4x}(6x+2)^{\frac{3}{2}}+9e^{4x}(6x+2)^{\frac{1}{2}}=e^{4x}(6x+2)^{\frac{1}{2}}\left(4\left(6x+2\right)+9\right)$ **Note: This mark is dependent on the first M Mark**
	A1	Correct $A = 24$ and correct $B = 17$
(b)	M1	Attempt at the quotient rule. Numerator must be the **difference** of two terms (either way round) of the form $k\cos 3x(2x-4)^3 - l\sin 3x(2x-4)^2$, $k = \pm 3$ $l \geqslant 3$ OR accept: (terms wrong way around) $l\sin 3x(2x-4)^2 - k\cos 3x(2x-4)^3$, $k = \pm 3$ $l \geqslant 3$ Denominator must be of the form $(2x-4)^6$ or accept $\left[(2x-4)^3\right]^2$
	A1	Either term correct, either way round.
	A1	Fully correct
	ALT	
	M1	Use of product rule to give an expression of the form $k\cos 3x(2x-4)^{-3} - l\sin 3x(2x-4)^{-4}$, $k = \pm 3$ $l \geqslant 3$ Accept for this mark $l\sin 3x(2x-4)^{-4} - k\cos 3x(2x-4)^{-3}$
	A1	Either term correct.
	A1	Fully correct
	NB: Ignore any subsequent simplification of their derivative.	

Type MS · The quotient rule: differentiating fractions

Answer 8 of 16 · Question 5 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 9 of 16 · Question 2 part whole · 5 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

2	$\frac{dy}{dx} = \sin 2x \times \frac{1}{2} \times 2 \times (3+2x)^{-\frac{1}{2}} + (3+2x)^{\frac{1}{2}} \times 2 \times \cos 2x$	M1A1A1
	$\frac{dy}{dx} = \frac{\sin 2x}{\sqrt{3+2x}} + 2 \times \sqrt{(3+2x)} \cos 2x = \frac{\sin 2x + 2(3+2x)\cos 2x}{\sqrt{3+2x}} = ...$	dM1
	$\frac{dy}{dx} = \frac{\sin 2x + (6+4x)\cos 2x}{\sqrt{3+2x}}$ *	A1 cso
	Where $A = 6$ and $B = 4$	[5]
		Total 5 marks

Type MS · The quotient rule: differentiating fractions

Answer 9 of 16 · Question 2 part whole · 5 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Notes
M1	For an attempt to apply product rule: An attempt is as follows. • The formula used must be correct. $\frac{dy}{dx} = uv' + vu'$ • Minimally acceptable differentiation used within a correct formula. $\sin 2x \to k\cos 2x$ and $\sqrt{3+2x} \to l(3+2x)^{-\frac{1}{2}}$ where k and l are a positive integers and $k, l \neq 0$ (You will see $l = 1$ without other working which is correct)
A1	One term correct – (simplification not required) and the other minimally acceptable. Either $\sin 2x \times \frac{1}{2} \times 2 \times (3+2x)^{-\frac{1}{2}}$ Or $(3+2x)^{\frac{1}{2}} \times 2 \times \cos 2x$
A1	Fully correct – simplification not required $\frac{dy}{dx} = \sin 2x \times \frac{1}{2} \times 2 \times (3+2x)^{-\frac{1}{2}} + (3+2x)^{\frac{1}{2}} \times 2 \times \cos 2x$
dM1	For **correct** use of $\sqrt{3+2x}$ or $(3+2x)^{\frac{1}{2}}$ as a common denominator. **NB:** This mark is dependent on the first M mark scored.
A1 cso	For the correct $\frac{dy}{dx}$ as shown with no erroneous or missing working. Allow embedded values. E.g., $\frac{dy}{dx} = \frac{\sin 2x + (6+4x)\cos 2x}{\sqrt{3+2x}}$ is dM1A1

Type MS · The quotient rule: differentiating fractions

Answer 9 of 16 · Question 2 part whole · 5 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 10 of 16 · Question 4 part whole · 5 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

4	$\frac{dy}{dx} = \frac{2\cos 2x\left(x^2-9\right)^{\frac{1}{2}} - \frac{1}{2}\times 2x\sin 2x\left(x^2-9\right)^{-\frac{1}{2}}}{\left(x^2-9\right)}$	M1A1A1
	$\left\{\frac{dy}{dx} = \frac{\frac{2\cos 2x\left(x^2-9\right)^{\frac{1}{2}}\left(x^2-9\right)^{\frac{1}{2}} - \frac{1}{2}\times 2x\times\sin 2x}{\left(x^2-9\right)^{\frac{1}{2}}}}{\left(x^2-9\right)}\right\}$	
	$\frac{dy}{dx} = \frac{2\left(x^2-9\right)\cos 2x - x\sin 2x}{\sqrt{\left(x^2-9\right)^3}}$ *	dM1A1 cso [5]
	ALT	
	$y = \sin(2x)\,(x^2-9)^{-\frac{1}{2}}$ $\frac{dy}{dx} = 2\cos(2x)(x^2-9)^{-\frac{1}{2}} + \sin(2x)\left(-\frac{1}{2}\right)(2x)(x^2-9)^{-\frac{3}{2}}$	[M1A1A1
	$\left\{\frac{dy}{dx} = \frac{2\cos(2x)(x^2-9) - x\sin(2x)}{(x^2-9)^{\frac{3}{2}}}\right\}$	
	$\frac{dy}{dx} = \frac{2(x^2-9)\cos(2x) - x\sin(2x)}{\sqrt{(x^2-9)^3}}$ *	dM1A1 cso]
		Total 5 marks

Type MS · The quotient rule: differentiating fractions

Answer 10 of 16 · Question 4 part whole · 5 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

dM1	For an attempt to rearrange to the given form. Dependent on M1 being scored. An attempt requires obtaining a single fraction and multiplying numerator and denominator by $(x^2-9)^{\frac{1}{2}}$ (see { } in mark scheme).
A1 **cso**	Fully correct method to show $\frac{dy}{dx} = \frac{2(x^2-9)\cos 2x - x\sin 2x}{\sqrt{(x^2-9)^3}}$ Allow with $\sqrt{(x^2-9)^3}$ given as $(x^2-9)^{\frac{3}{2}}$ or $\sqrt{(x^2-9)}^3$
ALT – product rule	
M1	For an attempt at Product rule. The definition of an attempt is that there must be a correct attempt to differentiate at least one term. Attempt at differentiation of the terms: $\sin(2x)(x^2-9)^{-\frac{1}{2}} \rightarrow k\cos(2x)(x^2-9)^{-\frac{1}{2}}$ where k is an integer $\sin(2x)(x^2-9)^{-\frac{1}{2}} \rightarrow lx\sin(2x)(x^2-9)^{-\frac{3}{2}}$
A1	For correct differentiation of at least one term. Either $2\cos(2x)(x^2-9)^{-\frac{1}{2}}$ or $\sin(2x)\left(-\frac{1}{2}\right)(2x)(x^2-9)^{-\frac{3}{2}}$
A1	For a fully correct Product rule $\frac{dy}{dx} = 2\cos(2x)(x^2-9)^{-\frac{1}{2}} + \sin(2x)\left(-\frac{1}{2}\right)(2x)(x^2-9)^{-\frac{3}{2}}$
dM1	For an attempt to rearrange to the given form. Dependent on M1 being scored. An attempt requires obtaining a single fraction and multiplying numerator and denominator by $(x^2-9)^{\frac{3}{2}}$ (see { } in mark scheme).
A1 **cso**	Fully correct method to show $\frac{dy}{dx} = \frac{2(x^2-9)\cos 2x - x\sin 2x}{\sqrt{(x^2-9)^3}}$ Allow with $\sqrt{(x^2-9)^3}$ given as $(x^2-9)^{\frac{3}{2}}$ or $\sqrt{(x^2-9)}^3$

Type MS · The quotient rule: differentiating fractions

Answer 10 of 16 · Question 4 part whole · 5 marks

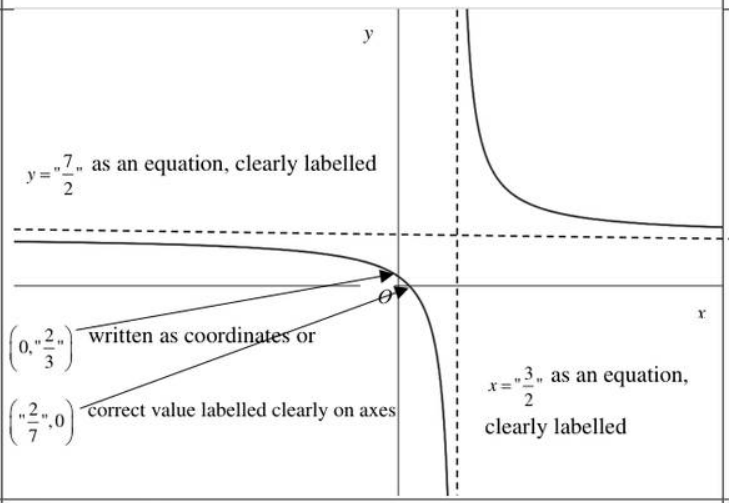
2021 NOV · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 11 of 16 · Question 10 part c · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 5 · find part c in this mark scheme extract

10 (a) (i)	$x = \frac{3}{2}$	B1
(ii)	$y = \frac{7}{2}$	B1 [2]
(b)	$\left(\frac{2}{7}, 0\right)$ $\left(0, \frac{2}{3}\right)$	B1 B1 [2]
(c)	$\frac{7(2x-3)-2(7x-2)}{(2x-3)^2}$	M1 A1
	$\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$	A1
	Correct conclusion	B1 [4]
	ALT – product rule	M1 A1
	$7(2x-3)^{-1} + (7x-2)(-1)(2)(2x-3)^{-2}$	
	$\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$	A1
	Correct conclusion	B1
(d)		B1 (curve) B1ft (asymptotes) B1ft (intersections with x- and y-axes) [3]
(e)	$-\frac{1}{17} = "\frac{-17}{(2x-3)^2}"$	M1
	"$(2x-3)^2 = 17^2$" **or** "$4x^2 - 12x - 280 = 0$" oe	dM1
	$x = 10$ $y = 4$ or $(10, 4)$	A1
	$y - "4" = 17(x - "10")$ **or** "4" = –17 × "10" + c leading to c =	M1
	$y = 17x - 166$ oe	A1

Type MS · The quotient rule: differentiating fractions

Answer 11 of 16 · Question 10 part c · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 5 · find part c in this mark scheme extract

	"$17x - 166" = \dfrac{7x-2}{2x-3}$ → $17x^2 - 195x + 250$	M1
	$x = \dfrac{25}{17}$ $y = -141$ or $\left(\frac{25}{17}, -141\right)$	A1 [7]
		Total 18 marks

Type MS · The quotient rule: differentiating fractions

Answer 11 of 16 · Question 10 part c · 4 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 5 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
		If a candidate gives **no response** to (a) and/or (b) but shows the correct answers on the graph we will award the marks. Where answers are given in (a) and/or (b) these should be marked as they stand with no reference to the graph. Ignore labelling of (i) and (ii) and mark (a) together.
(a)(i)	B1	For $x = \frac{3}{2}$ oe
(a)(ii)	B1	For $y = \frac{7}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x = \frac{2}{7}$ and/or $y = \frac{2}{3}$ given
(c)	M1	Attempt the quotient rule. Numerator must be the difference of two terms (either way round) of the form $A.(2x-3) - B.(7x-2)$, A and $B > 1$. Denominator must be of the form $(2x-3)^2$
	A1	Either term on the numerator correct (either way round), dependent on previous method mark.
	A1	Obtains $\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$
	B1	Correct conclusion based on correct working only, for example, (the numerator is a negative number and) the denominator is always positive and therefore the fraction/gradient is always negative.
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x-3)^{-1} + d(7x-2)(2x-3)^{-2}$ for constants c, d.
	A1	Either term correct, dependent on previous method mark.
	A1	Obtains $\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$
	B1	Correct conclusion based on correct working only, for example, (the numerator is a negative number and) the denominator is always positive and therefore the fraction/gradient is always negative.
(d)	B1	Two branches drawn in the correct two "quadrants" created by the two aymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes. Allow BOD if intention is for curve to run alongside asymptote but there is a slight deviation back on itself.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(e)	M1	Sets their differentiated function from part (c) = $-\frac{1}{17}$
	dM1	Rearranges to get to an equation of the form shown with no denominators or a 3TQ and solves using an acceptable method to obtain $x = \ldots$. Dependent on previous method mark
	A1	Correct values for point A (10, 4)
	M1	Uses their values for x and y (from an attempt at working with gradient of the curve) with gradient 17 to find an equation for l (if using $y = mx + c$, must be a complete method arriving at $c =$) If correct $c = -166$.
	A1	Correct equation, any form

Type MS · The quotient rule: differentiating fractions

Answer 11 of 16 · Question 10 part c · 4 marks

2022 JUN · Paper 02 (regional R) · extract 4 of 5 · find part c in this mark scheme extract

	M1	Sets their equation for the normal equal to the curve, makes a **correct** rearrangement to remove any denominator and forms a 3TQ Note this method mark is not dependant.
	A1	Correct exact values for x and y

Type MS · The quotient rule: differentiating fractions

Answer 11 of 16 · Question 10 part c · 4 marks

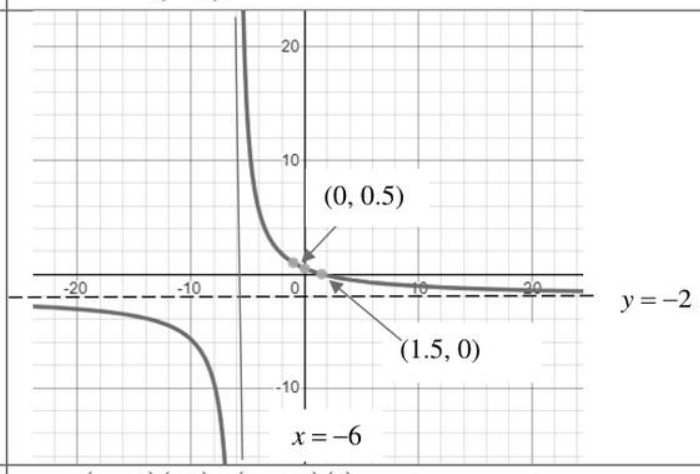
2022 JUN · Paper 02 (regional R) · extract 5 of 5 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 12 of 16 · Question 9 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 4 · find part d in this mark scheme extract

9(a)	(i) $y=-2$ (ii) $x=-6$	B1 B1 [2]
(b)	(i) $\left(\frac{3}{2},0\right)$ (ii) $\left(0,\frac{1}{2}\right)$	B1 B1 [2]
(c)		B1 – shape B1ft– Asymptotes B1ft – Intersections [3]
(d)	$\frac{dy}{dx}=\frac{(x+6)(-2)-(3-2x)(1)}{(x+6)^2}$ $\frac{dy}{dx}=\frac{-15}{(x+6)^2}$ with conclusion; numerator negative, denominator always positive, $\frac{-}{+}\Rightarrow$ negative	M1A1 A1 [3]
(e)	$\frac{-15}{(x+6)^2}=-\frac{3}{5}\Rightarrow 25=(x+6)^2\Rightarrow x=-6\pm5=-11,\quad -1$ $y=\frac{3-2(-1)}{-1+6}=1$ $y-1=-\frac{3}{5}(x-[-1])\Rightarrow y=-\frac{3}{5}x+\frac{2}{5}\Rightarrow k=\frac{2}{5}$	M1A1 B1 M1A1 [5]
		Total 15 marks

Type MS · The quotient rule: differentiating fractions

Answer 12 of 16 · Question 9 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 4 · find part d in this mark scheme extract

Part	Mark	Notes
(a) (i) (ii)	B1	For the correct equation. If their equations are not labelled (i) and (ii) accept them in the order given only. For example, the following presentation is B0B0 $x=-6$ $y=-2$
	B1	For the correct equation.
(b)		If their coordinates are not labelled (i) and (ii) or are given in the incorrect place [for example, (b)(i) $y=\frac{1}{2}$ is B0] accept them in the order given only. For example, the following presentation is B0B0 $\left(0,\frac{1}{2}\right)$ $\left(\frac{3}{2},0\right)$
(i)	B1	For the correct coordinates. Accept $x=\frac{3}{2}$
(ii)	B1	For the correct coordinates. Accept $y=\frac{1}{2}$
(c)	B1	For the correct shape with two branches the correct way around anywhere in the grid. It must be asymptotic in nature. The ends of the curve must not come back on themselves. Whilst you need to be fairly generous, any obvious turning back is B0
	B1ft	For the correct asymptotes drawn and labelled with at least one branch of the curve [which must be asymptotic in nature in at least one branch] in the correct place seen. Ft their asymptotes Accept the vertical line drawn shown passing through their -6 and the horizontal line drawn shown passing through their -2
	B1ft	For the correct coordinates of intersections with the relevant branch of the curve drawn correctly seen. The curve must go through the points. Do not accept touching the axis. Allow $\frac{1}{2}$ marked on y-axis and $\frac{3}{2}$ marked on the x-axis. Ft their coordinates of intersections.
(d)	M1	For an attempt at quotient rule. • The denominator must be squared, or accept $(x+6)(x+6)$ or $x^2+12x+36$ • Both $(x+6)$ and $(3-2x)$ differentiated correctly. • The two terms in the numerator subtracted either way around.
	A1	Fully correct derivative (simplification not required for this mark).
	A1	For a correct simplified derivative with a correct conclusion. For example: $(x+6)^2 \geqslant 0$, -15 is negative, $\frac{\text{negative}}{\text{positive}}$ is always negative. [Accept also $(x+6)^2>0$]

Type MS · The quotient rule: differentiating fractions

Answer 12 of 16 · Question 9 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 4 · find part d in this mark scheme extract

(e)	M1	For setting the value of $-\frac{3}{5}$ = their $\frac{dy}{dx}$ with an attempt to find at least one value of x. Allow this even if their derivative results in a linear equation.
	A1	For **both** correct values of x
	B1	For $y=1$ using $x=-1$ OR For $y=-5$ using $x=-11$
	M1	For forming an equation of the line with either $x=-1,\quad y=1$ or $x=-11,\quad y=-5$ or their x and their y
	A1	For the correct value of k (accept an embedded value). $y=-\frac{3}{5}x+\frac{2}{5}\Rightarrow k=\frac{2}{5}$ You can award this mark even if the previous A mark has not been scored. So, for a correct solution without showing that $x=-11$ score M1A0B1M1A1
	ALT for last 3 marks	
	B1	Sets $-\frac{3}{5}x+k=\frac{3-2x}{x+6}$
	M1	Substitutes $x=-1$ or -11 into the above equation
	A1	For $k=\frac{2}{5}$ You can award this mark even if the previous A mark has not been scored. So, for a correct solution without showing that $x=-11$ score M1A0B1M1A1

Type MS · The quotient rule: differentiating fractions

Answer 12 of 16 · Question 9 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 4 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 13 of 16 · Question 11 part a · 3 marks

2022 JUN · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

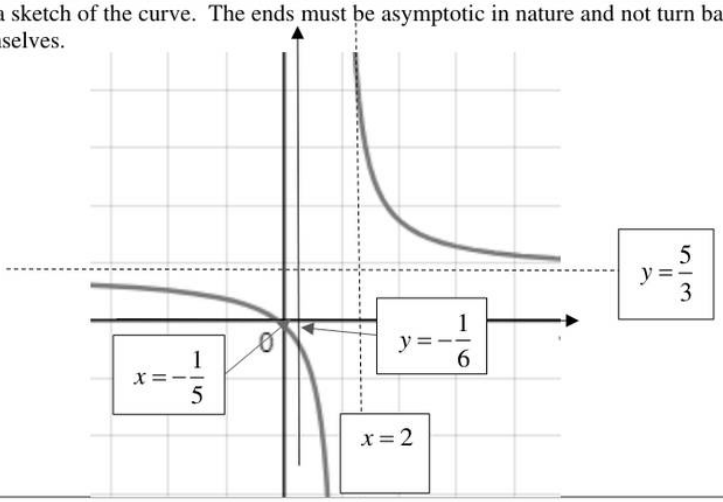
11 a	$\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$	M1A1A1 (3)
b (i)	At $x = 0$ $\frac{dy}{dx} = -\frac{11}{12}$	B1
	So $\frac{dy}{dx} = \frac{-6(2a-1)-a}{36} = -\frac{11}{12}$	M1
	$-13a+6 = -33 \Rightarrow a = 3$*	dM1 A1 cso (4)
b (ii)	$y = \frac{5x+1}{3x-6}$*	B1 cso (1)
c	Asymptote with equation drawn at $y = 2$	B1
	Asymptote with equation drawn at $x = \frac{5}{3}$	B1
	Correct curve drawn with two branches	B1
	$-\frac{1}{5}$ and $-\frac{1}{6}$ labelled on x and y axes	B1 B1 (5)
d	$\frac{12}{11}x - \frac{1}{6} = \frac{5x+1}{3x-6}$	M1
	$(3x-6)(72x-11) = 66(5x+1)$	M1
	$x(72x-265) = 0$ oe	M1
	$x = \frac{265}{72}$ oe	A1 (4)
		Total 17 marks

Part	Mark	Notes
(a)	**M1**	For attempting to use a correct quotient rule: The definition of an attempt is as follows: • The denominator must be correct • The terms in the numerator must be subtracted either way around • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \to \pm(2a-1)$ and $ax-6 \to \pm a$
	A1	For one term in the numerator correct
	A1	For a fully correct differentiated expression. $\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$ No simplification is required.

Type MS · The quotient rule: differentiating fractions

Answer 13 of 16 · Question 11 part a · 3 marks

2022 JUN · Paper 02 · extract 2 of 4 · find part a in this mark scheme extract

	Uses Product Rule	
	M1	For attempting to use a correct product rule: The definition of an attempt is as follows: • The terms in the numerator must be added. • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \rightarrow \pm(2a-1)$ and $(ax-6)^{-1} \rightarrow \pm a(ax-6)^{-2}$
	A1	For one term correct
	A1	For a fully correct differentiation $\frac{dy}{dx} = \left[(2a-1)x+1\right](a)(ax-6)^{-2} + (ax-6)^{-1}(2a-1)$
(b)(i)	**B1**	For finding the gradient of the tangent $= -\frac{11}{12}$
	M1	For using that at the point A $x = 0$ and setting their $\frac{dy}{dx} = \left(-\frac{11}{12}\right)$ which must be the gradient of their tangent to obtain $\frac{-6(2a-1)-a}{36} = '-\frac{11}{12}' \Rightarrow \left(\frac{-13a+6}{36} = -\frac{11}{12}\right)$
	dM1	For simplifying to a linear equation and attempting to solve.
	A1	For $a = 3$ with no errors seen
(b)(ii)	**B1**	For $y = \frac{5x+1}{3x-6}$ exactly in that form. You must see evidence of substitution of $a = 3$ for this mark. There must be $y =$ …. [Condone missing $x \neq 2$]
(c)		This is a sketch of the curve. The ends must be asymptotic in nature and not turn back on themselves. 
	B1	For asymptote **with equation drawn** at $y = \frac{5}{3}$ OR $x = 2$
	B1	For both asymptotes **with equations drawn** at $y = \frac{5}{3}$ AND $x = 2$
	B1	Correct curve drawn with two branches with one branch in 1st Quadrant and another branch in Quadrants 2, 3 and 4 with the graph drawn the correct way around. The graph must be asymptotic in nature.

Type MS · The quotient rule: differentiating fractions

Answer 13 of 16 · Question 11 part a · 3 marks

2022 JUN · Paper 02 · extract 3 of 4 · find part a in this mark scheme extract

	B1	$-\frac{1}{5}$ labelled on x-axis **or** $-\frac{1}{6}$ labelled on y- axis
	B1	$-\frac{1}{5}$ labelled on x axis **and** $-\frac{1}{6}$ labelled on y- axis
(d)	**M1**	For setting $\frac{12}{11}x-\frac{1}{6}=\frac{5x+1}{3x-6}$ [eliminating y from both equations]
	M1	For forming a quadratic equation, 2TQ or 3TQ in any order. $(3x-6)(72x-11)=66(5x+1) \Rightarrow 72x^2-265x=0$
	M1	For solving their QE provided it is either a 2TQ or a 3TQ
	A1	For $x=\frac{265}{72}$ oe

Type MS · The quotient rule: differentiating fractions

Answer 13 of 16 · Question 11 part a · 3 marks

2022 JUN · Paper 02 · extract 4 of 4 · find part a in this mark scheme extract

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Type MS · The quotient rule: differentiating fractions

Answer 14 of 16 · Question 3 part whole · 5 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

3	$4\times2(2x-1)^3e^{3x}+3e^{3x}(2x-1)^4$ *oe* $4\times2(2(1)-1)^3e^3+3e^3(2(1)-1)^4$ $11e^3$ cao	M1A1A1 dM1 A1
		Total 5 marks

Type MS · The quotient rule: differentiating fractions

Answer 14 of 16 · Question 3 part whole · 5 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 15 of 16 · Question 6 part a · 5 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
6(a)	$\frac{dy}{dx}=\frac{\left(x^2+1\right)2xe^{\left(x^2+1\right)}-2xe^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$	M1A1A1
	$\frac{dy}{dx}=\frac{2xe^{\left(x^2+1\right)}\left(x^2+1-1\right)}{\left(x^2+1\right)^2}=\frac{2x^3e^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$	M1A1 cso [5]
(b)	When $x=-1$ $\frac{dy}{dx}=\frac{-e^2}{2}$, $y=\frac{e^2}{2}$	B1ft, B1
	$y-\frac{e^2}{2}=-\frac{e^2}{2}(x+1),\Rightarrow y=-\frac{e^2x}{2}$ oe	M1A1ft, A1 [5]
		Total 10 marks

Question	Notes	Marks
6(a)	$y=\frac{e^{\left(x^2+1\right)}}{x^2+1}$	
	Using Quotient Rule $\frac{dy}{dx}=\frac{\left(x^2+1\right)2xe^{\left(x^2+1\right)}-2xe^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$ • For an attempt to differentiate both $e^{\left(x^2+1\right)}$ and x^2+1 Award for either $e^{x^2+1}\Rightarrow 2xe^{x^2+1}$ or $x^2+1\Rightarrow 2x$ but both must be **changed** expressions. • Numerator is to have two terms in either order **subtracted.** • Denominator must be $\left(x^2+1\right)^2$	M1
	At least one term fully correct in the numerator	A1
	Fully correct unsimplified.	A1
	For an attempt to take out a common factor of either $2x$ or $e^{\left(x^2+1\right)}$ $\frac{dy}{dx}=\frac{2xe^{\left(x^2+1\right)}\left(x^2+1-1\right)}{\left(x^2+1\right)^2}$ OR Multiplies out the first term in the numerator $\frac{2x^3e^{\left(x^2+1\right)}+2xe^{\left(x^2+1\right)}-2xe^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$	M1 [M1]

Type MS · The quotient rule: differentiating fractions

Answer 15 of 16 · Question 6 part a · 5 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

	Using Product Rule	
	$\frac{dy}{dx} = \left(x^2+1\right)^{-1} \times 2xe^{x^2+1} + (-2x)e^{x^2+1} \times \left(x^2+1\right)^{-2}$ • For an attempt to differentiate both $e^{\left(x^2+1\right)}$ and $\left(x^2+1\right)^{-1}$ Award for either $e^{x^2+1} \Rightarrow 2xe^{x^2+1}$ or $\left(x^2+1\right)^{-1} \Rightarrow (-2x)\left(x^2+1\right)^{-2}$ but both must be **changed** expressions. • Numerator is to have two terms added	M1
	At least one term correct	A1
	Fully correct unsimplified or simplified	A1
	For an attempt to take out a common factor of either $2x$ or $e^{\left(x^2+1\right)}$ and set a common denominator of $\left(x^2+1\right)^2$	M1
	For the fully correct expression for the derivative, $\frac{dy}{dx} = \frac{2x^3e^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$ NB: Any further simplification [e.g., cancelling $\left(x^2+1\right)$] following a correct answer seen, is A0.	A1 [5]
(b)	When $x=-1$, $\frac{dy}{dx} = \frac{-e^2}{2}$ ft their K in their $\frac{dy}{dx}$ Allow awrt $\frac{dy}{dx} = -3.7$ Even allow $\frac{dy}{dx} = \frac{-Ke^2}{4}$	B1ft
	When $x=-1$, $y = \frac{e^2}{2}$ Allow awrt $y = 3.7$	B1
	For a correctly used method for the equation of the tangent using their values for $\frac{dy}{dx}$ and y $y - '\frac{e^2}{2}' = '-\frac{e^2}{2}'(x+1)$ Also allow: $y-3.7 = -3.7(x+1)$ If they use $y = mx+c$ they must obtain a value for c for the award of this mark.	M1
	For the correct equation in any form	A1ft
	For a correct simplified equation in any form but this must be in exact form. $y = -\frac{e^2x}{2}$ or $2y + e^2x = 0$	A1 [5]
		Total 10 marks

Type MS · The quotient rule: differentiating fractions

Answer 15 of 16 · Question 6 part a · 5 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · The quotient rule: differentiating fractions

Answer 16 of 16 · Question 8 part a · 5 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

8(a)	$\left(\frac{dy}{dx}=\right)\frac{5x^2\times2\times3\times e^{3x+1}-10x\times2e^{3x+1}}{(5x^2)^2}$ oe	M1A1A1
	$\left(\frac{dy}{dx}=\right)\frac{xe^{3x+1}(30x-20)}{25x^4}$ or $\frac{10xe^{3x+1}(3x-2)}{25x^4}$	dM1
	$=\left[\frac{10e^{3x+1}(3x-2)}{25x^3}\right]=\frac{2e^{3x+1}(3x-2)}{5x^3}$	A1 [5]
ALT – use of product rule	$\left(\frac{dy}{dx}=\right)2e^{3x+1}(-1)\times2\times5x\times(5x^2)^{-2}+2\times3e^{3x+1}(5x^2)^{-1}$ oe	M1A1A1
	$=x\times(5x^2)^{-2}e^{3x+1}(-20+30x)$ or $10x\times(5x^2)^{-2}e^{3x+1}(-2+3x)$	dM1
	$=\left[\frac{10e^{3x+1}(3x-2)}{25x^3}\right]=\frac{2e^{3x+1}(3x-2)}{5x^3}$	A1 [5]
(b)	$\left(\frac{\delta y}{y}\times100=\right)"\frac{2xe^{3x+1}(3x-2)}{5x^4}"\times\delta x\times\frac{100}{y}$ or $"\frac{2e^{3x+1}(3x-2)}{5x^3}"\times\delta x\times\frac{100}{y}$ oe	M1
	or $(\delta y=)"\frac{2xe^{3x+1}(3x-2)}{5x^4}"\times\delta x$ or $"\frac{2e^{3x+1}(3x-2)}{5x^3}"\times\delta x$ or $y\times\frac{(3x-2)}{5x}\times\delta x$ δx may also be written as $0.02x$	
	$\left(\frac{\delta y}{y}\times100=\right)\frac{2e^{3x+1}(3x-2)}{5x^3}\times0.02x\times\frac{100}{\frac{2e^{3x+1}}{5x^2}}$ oe	dM1
	or $y\times\frac{(3x-2)}{x}\times0.02x\times\frac{100}{y}$ $(\%\text{ change in } y=)6x-4$	A1 [3]
		Total 8 marks

Type MS · The quotient rule: differentiating fractions

Answer 16 of 16 · Question 8 part a · 5 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For attempting to use the quotient rule. The definition of an attempt is as follows: • The denominator must be correct. • The terms in the numerator subtracted either way around The expression **must** be of the form: $\frac{K\times x^2\times e^{3x+1}-L\times x\times e^{3x+1}}{(5x^2)^2}$ $K\neq 10$ or 15 and $L\neq 10$ and $K,L>0$ Note K and L may not be simplified and will often need to be checked.
	A1	Following M1 (general principle of marking), at least one fully correct simplified or unsimplified term on the numerator.
	A1	Fully correct **unsimplified** derivative
	dM1	For correctly taking out common factors of x and e^{3x+1} and attempting to simplify. A numerical factor does not need to be taken out at this point. Dependent on the 1st method mark.
	A1	For the correct derivative in the required form.
ALT use of product rule	M1	For attempting to use the product rule to give an expression of the form $Me^{3x+1}x\times(5x^2)^{-2}+Ne^{3x+1}(5x^2)^{-1}$ $M<0$ and $\neq 4,10$ $N>0$ and $\neq 2,3$
	A1	Following M1 (general principle of marking), at least one term is fully correct.
	A1	Fully correct **unsimplified** derivative
	M1	For correctly taking out common factors of x and $(5x^2)^{-2}$ and e^{3x+1} and attempting to simplify. A numerical factor does not need to be taken out at this point. Dependent on the 1st method mark.
	A1	For the correct derivative in the required form.
For the final M mark in particular – any work felt to be of credit not fitting the mark scheme – send to Review.		
(b)	M1	For any of the statements shown, using their answer from part (a), which must be of the form: $\frac{Ae^{3x+1}(Bx-C)}{Dx^3}$ Note, we'll allow candidates to use an expression here that is not quite of the form given in part (a). This mark may be implied by further correct working.
	dM1	For substituting in the expression for y **and** a correct attempt to simplify their expression to the required form. You must check their working here, do not allow an answer that has come from incorrect working (general principle anyway, but extra care needed for this question). Condone 100 written as 100% if the candidate is correctly multiplying by 100. Dependent on previous method mark.
	A1	For the correct expression as shown.

Type MS · The quotient rule: differentiating fractions

Answer 16 of 16 · Question 8 part a · 5 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks